Boss Challenge — Paper 18 (Class 7)

Reproduction in Plants | Forests | Symmetry | Algebraic Expressions

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

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1. A plant that grows a new plant from a single parent, with no joining of cells, is reproducing by the method called:

- **(A)** germination
- **(B)** asexual reproduction
- **(C)** sexual reproduction
- **(D)** pollination

2. A line that divides a figure into two halves that are exact mirror images of each other is called a:

- **(A)** line of symmetry
- **(B)** line of best fit
- **(C)** diagonal of a square
- **(D)** number line

3. The branchy top of the tallest forest trees spread out so that they touch and form a green roof over the forest. This roof is called the:

- **(A)** canopy
- **(B)** understorey
- **(C)** crown
- **(D)** humus

4. In algebra, a letter such as x or n that can stand for different numbers is called a:

- **(A)** constant
- **(B)** coefficient
- **(C)** equation
- **(D)** variable

5. In yeast, a tiny bulge grows on the parent cell, gets bigger, and then breaks away as a new yeast. This way of reproducing is called:

- **(A)** budding
- **(B)** grafting
- **(C)** fragmentation
- **(D)** pollination

6. A square can be folded into two matching halves in several ways; in total, how many lines of symmetry does it have?

- **(A)** 0
- **(B)** 2
- **(C)** 4
- **(D)** 1

7. The branchy, leafy top part of a single tree is called its:

(A) canopy
(B) root
(C) crown
(D) herb

8. In the term 7x, the number 7 that multiplies the variable x is called the:

(A) exponent
(B) variable
(C) coefficient
(D) constant

9. A potato can grow a whole new plant from the small 'eyes' on its surface. Growing a new plant from a part of the plant body like this is called:

(A) vegetative propagation
(B) fertilisation
(C) seed germination
(D) pollination

10. A rectangle that is not a square has how many lines of symmetry?

(A) 0
(B) 1
(C) 4
(D) 2

11. In a forest, the dark-coloured material formed when dead leaves and animal remains rot and mix into the soil is called:

(A) humus
(B) crown
(C) sapling
(D) canopy

12. Which of the following pairs are LIKE terms (terms that can be added together directly)?

(A) 5x and 3x
(B) 5 and 3x
(C) 5x and $3x^2$
(D) 5x and 3y

13. In a flower, the part that makes pollen - the male part - is the:

(A) sepal
(B) stamen
(C) pistil
(D) petal

14. How many lines of symmetry does an equilateral triangle (all three sides equal) have?

- **(A)** 2
- **(B)** 3
- **(C)** 1
- **(D)** 0

15. The small living things in the soil that break down dead leaves and dead animals into simple substances are called:

- **(A)** predators
- **(B)** decomposers
- **(C)** herbivores
- **(D)** producers

16. Adding the like terms, $2x + 3x$ equals:

- **(A)** $5x$
- **(B)** $23x$
- **(C)** $5x^2$
- **(D)** $6x$

17. The female part of a flower, which has the stigma, style and ovary, is called the:

- **(A)** stamen
- **(B)** filament
- **(C)** anther
- **(D)** pistil

18. How many lines of symmetry does an isosceles triangle (only two sides equal) have?

- **(A)** 2
- **(B)** 3
- **(C)** 0
- **(D)** 1

19. Forests are often called the 'green lungs' of the Earth because they:

- **(A)** make the soil dry and hard
- **(B)** store water and release it as ice
- **(C)** release oxygen and take in carbon dioxide
- **(D)** release carbon dioxide and take in oxygen

20. Write 'a number x increased by 5' as an algebraic expression:

- **(A)** $5x$
- **(B)** $x + 5$
- **(C)** $x / 5$
- **(D)** $x - 5$

21. The transfer of pollen from the anther of a flower to the stigma of a flower is called:

- **(A)** germination
- **(B)** fertilisation
- **(C)** pollination
- **(D)** dispersal

22. A scalene triangle, with all three sides of different lengths, has how many lines of symmetry?

- **(A)** 2
- **(B)** 0
- **(C)** 1
- **(D)** 3

23. Why does the soil under a thick forest hardly get washed away even during heavy rain?

- **(A)** the soil in forests is made of stone
- **(B)** rain never falls inside forests
- **(C)** the roots of the trees hold the soil firmly in place
- **(D)** the canopy turns the rain into steam

24. Write 'twice a number n, then add 3' as an algebraic expression:

- **(A)** 2(n + 3)
- **(B)** n + 3
- **(C)** 2n + 3
- **(D)** 2 + 3n

25. When the pollen of a flower lands on the stigma of a different flower of the same kind, it is called:

- **(A)** vegetative propagation
- **(B)** fertilisation
- **(C)** self-pollination
- **(D)** cross-pollination

26. Through its centre, a round disc can be folded into matching halves in how many different ways - that is, its number of lines of symmetry is:

- **(A)** 0
- **(B)** only 4
- **(C)** only 2
- **(D)** an unlimited (infinite) number

27. A simple food chain in a forest could be: grass -> deer -> tiger. In this chain, the grass is the:

- **(A)** producer
- **(B)** consumer
- **(C)** decomposer
- **(D)** predator

28. In the term -4y, the coefficient of y (including its sign) is:

- **(A)** y
- **(B)** -4
- **(C)** 4
- **(D)** -4y

29. The joining (fusion) of the male cell from pollen with the female egg cell to form a single cell is called:

- **(A)** pollination
- **(B)** germination
- **(C)** budding
- **(D)** fertilisation

30. How many lines of symmetry does a regular pentagon (5 equal sides) have?

- **(A)** 5
- **(B)** 10
- **(C)** 4
- **(D)** 2

31. In the food chain grass -> deer -> tiger, the deer is best described as a:

- **(A)** plant that the grass feeds on
- **(B)** producer that makes its own food
- **(C)** herbivore that is eaten by the tiger
- **(D)** decomposer that rots dead leaves

32. An algebraic expression that has exactly one term, such as 7x, is called a:

- **(A)** equation
- **(B)** trinomial
- **(C)** binomial
- **(D)** monomial

33. After fertilisation, the single cell that is formed and which later grows into the embryo of a seed is the:

- **(A)** stigma
- **(B)** zygote
- **(C)** pollen
- **(D)** ovary

34. A regular hexagon has 6 equal sides. How many lines of symmetry does it have?

- **(A)** 6
- **(B)** 3
- **(C)** 12
- **(D)** 4

35. Many separate food chains in a forest cross and link with one another. This whole network of connected food chains is called a:

- **(A)** food chain
- **(B)** humus
- **(C)** canopy
- **(D)** food web

36. An expression with exactly two terms, such as $2x + 5$, is called a:

- **(A)** variable
- **(B)** binomial
- **(C)** trinomial
- **(D)** monomial

37. After fertilisation, the ovary of the flower grows and ripens into the:

- **(A)** fruit
- **(B)** flower
- **(C)** root
- **(D)** seed

38. For any regular polygon, the number of lines of symmetry is equal to:

- **(A)** always just 2
- **(B)** the number of its sides
- **(C)** twice the number of its sides
- **(D)** half the number of its sides

39. Forests help bring rain to a region mainly because the trees:

- **(A)** store rain inside their trunks
- **(B)** block the wind completely
- **(C)** give out water vapour from their leaves, which helps clouds form
- **(D)** burn and send smoke into the sky

40. What is the value of the expression $3x + 2$ when $x = 4$?

- **(A)** 9
- **(B)** 14
- **(C)** 18
- **(D)** 20

41. A seed grows from which part of the flower after fertilisation?

- **(A)** the anther
- **(B)** the petal
- **(C)** the sepal
- **(D)** the ovule

42. The capital letter A (block form) has how many lines of symmetry?

(A) 2 lines
(B) 1 horizontal line
(C) 1 (a vertical line down the middle)
(D) 0 lines

43. Why does the forest floor not get buried under huge piles of dead leaves year after year?

(A) decomposers keep rotting the dead leaves into humus
(B) dead leaves dissolve in rain water
(C) animals eat all the fallen leaves
(D) the wind blows every dead leaf away

44. In the expression $5x + 8$, the term 8, which has no variable, is called the:

(A) like term
(B) variable
(C) coefficient
(D) constant term

45. Light seeds with hair or wings, such as those of the drumstick or the cotton plant, are usually carried away by:

(A) explosion
(B) wind
(C) heat
(D) water

46. The capital letter H (block form) has how many lines of symmetry?

(A) 2 (one vertical and one horizontal)
(B) 4
(C) 1
(D) 0

47. A young tree, still growing toward its full size, is called a:

(A) humus
(B) herb
(C) sapling
(D) canopy

48. Subtracting like terms, $9a - 4a$ equals:

(A) 5
(B) 13a
(C) $5a^2$
(D) 5a

49. The thick, fibrous husk of a coconut traps air and lets the seed float, so the coconut is spread mainly by:

- **(A)** insects
- **(B)** water
- **(C)** birds
- **(D)** wind

50. Turning a square about its centre, the smallest angle of turn that makes it look exactly the same as before is:

- **(A)** 360°
- **(B)** 90°
- **(C)** 45°
- **(D)** 180°

51. Forests are described as a renewable resource. This means that:

- **(A)** if used wisely, forests can grow back and renew themselves
- **(B)** forests can never be used up no matter what
- **(C)** forests are made by factories when needed
- **(D)** once cut, a forest can never come back

52. Write 'the number of legs on c cows' as an algebraic expression (each cow has 4 legs):

- **(A)** c + 4
- **(B)** 4 + c
- **(C)** 4c
- **(D)** c / 4

53. Seeds of plants such as Xanthium and Urena have tiny hooks or spines so that they can be spread by:

- **(A)** being blown by wind
- **(B)** floating on water
- **(C)** clinging to the fur of animals
- **(D)** bursting from a pod

54. An equilateral triangle has rotational symmetry of order:

- **(A)** 6
- **(B)** 2
- **(C)** 3
- **(D)** 1

55. The layer of short, shade-loving trees and shrubs that grows below the tall canopy trees is called the:

- **(A)** canopy
- **(B)** crown
- **(C)** understorey
- **(D)** humus

56. The perimeter of a square with side of length s can be written as the expression:

- **(A)** 2s
- **(B)** s^2
- **(C)** 4s
- **(D)** s + 4

57. Why is it useful for a plant to have its seeds scattered far away rather than dropping them all at its own base?

- **(A)** the seeds become heavier as they travel
- **(B)** the new plants avoid crowding and competing with the parent for light and water
- **(C)** the parent plant can then move to a new place
- **(D)** it makes the flowers more colourful

58. What is the order of rotational symmetry of a rectangle that is not a square?

- **(A)** 1
- **(B)** 2
- **(C)** 3
- **(D)** 4

59. Cutting down large areas of forest, called deforestation, can lead to:

- **(A)** more soil erosion and floods
- **(B)** richer, deeper soil
- **(C)** more rainfall every year
- **(D)** cleaner air and more oxygen

60. Which of these is a TRINOMIAL (an expression with exactly three terms)?

- **(A)** 7x
- **(B)** 5
- **(C)** $x^2 + 2x + 1$
- **(D)** 2x + 1

61. Mosses and ferns do not make seeds. Instead they reproduce using tiny structures called:

- **(A)** spores
- **(B)** seeds
- **(C)** tubers
- **(D)** buds

62. A figure that looks the same only after a full 360° turn is said to have rotational symmetry of order:

- **(A)** 1
- **(B)** 0
- **(C)** 2
- **(D)** 4

63. In a forest, the nutrients of a dead tree are not lost forever because:

(A) animals carry all the nutrients out of the forest
(B) the dead tree turns straight into a new tree
(C) the nutrients evaporate into the air and are gone
(D) decomposers return them to the soil for new plants to use

64. Adding the expressions $(2x + 3)$ and $(4x + 5)$ gives:

(A) $6x^2 + 8$
(B) $6x + 8$
(C) $6x + 15$
(D) $8x + 8$

65. A flower that has both the stamen (male part) and the pistil (female part), like the mustard flower, is called a:

(A) bisexual flower
(B) unisexual flower
(C) compound flower
(D) artificial flower

66. A parallelogram (slanted, not a rectangle) has how many lines of symmetry?

(A) 1
(B) 2
(C) 0
(D) 4

67. Which group of living things in a forest can make its own food and so supports all the animals?

(A) the decomposers
(B) the carnivores
(C) the herbivores
(D) the green plants

68. Subtracting $(x + 2)$ from $(5x + 7)$ gives:

(A) $4x + 5$
(B) $5x + 5$
(C) $6x + 9$
(D) $4x + 9$

69. The stigma at the top of the pistil is usually sticky. The most useful reason for this stickiness is that it:

(A) helps pollen grains stick to it during pollination
(B) traps insects to eat them
(C) stores water for the plant
(D) makes the flower smell sweet

70. Which of these shapes has an unlimited (infinite) number of lines of symmetry?

(A) an equilateral triangle
(B) a circle
(C) a rectangle
(D) a square

71. Forests help refill the underground water (groundwater) because:

(A) trees pump water up from the rocks
(B) water sinks slowly into the soil instead of running off quickly
(C) the canopy turns rain into groundwater directly
(D) forests stop all rain from reaching the ground

72. A forest department plants saplings in rows, with exactly 12 saplings in each row. The number of saplings in r rows is best written as:

(A) r / 12
(B) 12r
(C) 12 + r
(D) 12 - r

73. Bryophyllum can grow tiny new plantlets along the notches at the edge of its leaf. This is an example of reproduction from a:

(A) root
(B) leaf
(C) flower
(D) seed

74. A leaf is folded along its central vein (the midrib) and the two halves match exactly. The midrib is acting as the leaf's:

(A) line of symmetry
(B) axis of rotation
(C) number line
(D) line of best fit

75. Which of the following is a benefit a forest provides to people and wildlife?

(A) dust storms and dry weather
(B) only firewood and nothing else
(C) noise and pollution
(D) wood, medicines, clean air and homes for many animals

76. A nursery sells each plant for ■15. Adding a fixed delivery charge of ■40, the total cost for p plants is:

(A) 55p
(B) 15 + 40p
(C) 15p + 40
(D) 15p - 40

77. Rose and sugarcane plants are often grown by cutting a piece of the stem and planting it. This method is called growing from a:

- **(A)** spore
- **(B)** seed
- **(C)** stem cutting
- **(D)** bud of a flower

78. A regular flower has 6 identical petals spaced evenly around its centre. How many lines of symmetry does this flower pattern have?

- **(A)** 6
- **(B)** 3
- **(C)** 1
- **(D)** 12

79. Insects and birds visiting forest flowers help the forest grow by:

- **(A)** eating all the forest's seeds
- **(B)** carrying pollen from flower to flower
- **(C)** drying up the forest soil
- **(D)** cutting down old trees

80. In the term 6mn, the factors (the parts multiplied together) are:

- **(A)** 6 and mn only
- **(B)** only 6
- **(C)** 6, m and n
- **(D)** 6 + m + n

81. In which case is the offspring exactly like the single parent, with no mixing of features from two parents?

- **(A)** asexual reproduction
- **(B)** sexual reproduction
- **(C)** fertilisation by two flowers
- **(D)** cross-pollination

82. The capital letter S (block form) has no line of symmetry, but turning it 180° gives the same shape. Its rotational symmetry has order:

- **(A)** 0
- **(B)** 1
- **(C)** 4
- **(D)** 2

83. Animals that eat fruits and then drop the seeds far away help the forest by:

- **(A)** eating the roots of grown trees
- **(B)** keeping all the seeds in one place
- **(C)** stopping new trees from growing
- **(D)** spreading the seeds to new places where they can grow

84. Which statement about the expression $4x + 9$ is correct?

- **(A)** x is the constant term
- **(B)** both 4 and 9 are coefficients of x
- **(C)** 4 is the coefficient of x and 9 is the constant term
- **(D)** 9 is the coefficient of x and 4 is the constant

85. Which of these is the correct order of events that leads to a seed?

- **(A)** seed formation -> pollination -> fertilisation
- **(B)** fertilisation -> pollination -> seed formation
- **(C)** pollination -> seed formation -> fertilisation
- **(D)** pollination -> fertilisation -> seed formation

86. A regular octagon (a stop-sign shape) has 8 equal sides. How many lines of symmetry does it have?

- **(A)** 16
- **(B)** 8
- **(C)** 4
- **(D)** 6

87. Which statement about the layers of a forest is correct?

- **(A)** the canopy is at the top, with the understorey and forest-floor herbs below it
- **(B)** the understorey is the highest layer
- **(C)** the herbs grow above the canopy
- **(D)** all the layers are exactly the same height

88. What is the value of the expression $2n - 1$ when $n = 6$?

- **(A)** 12
- **(B)** 10
- **(C)** 13
- **(D)** 11

89. Some water plants like Spirogyra simply break into two or more pieces, and each piece grows into a new plant. This is called:

- **(A)** budding
- **(B)** pollination
- **(C)** grafting
- **(D)** fragmentation

90. A rhombus (a 'diamond' with all four sides equal but not a square) has its lines of symmetry along its:

- **(A)** centre point only
- **(B)** four sides at once
- **(C)** midpoints of opposite sides
- **(D)** two diagonals

91. Forests are important for keeping the air's gases balanced because they:

(A) use up carbon dioxide and add oxygen during the day
(B) use up oxygen and add carbon dioxide all day
(C) remove all gases from the air
(D) turn carbon dioxide into stone

92. In the number pattern 5, 10, 15, 20, … each term is 5 times its position. The expression for the nth term is:

(A) n - 5
(B) 5 + n
(C) n + 5
(D) 5n

93. Why are the petals of many flowers brightly coloured and scented?

(A) to protect the flower from rain
(B) to store seeds until they ripen
(C) to attract insects and birds that carry pollen
(D) to make food for the plant by photosynthesis

94. Reflection symmetry of a figure is exactly the same idea as the figure having a:

(A) number of sides
(B) line of symmetry
(C) rotational symmetry
(D) centre of the figure

95. The best way to protect a forest while still using its wood is to:

(A) cut down the whole forest at once
(B) never use any wood from forests at all
(C) burn the old trees to make room
(D) cut trees carefully and plant new ones to replace them

96. Simplifying 3x + 2y + 5x - y by collecting like terms gives:

(A) 8x + 3y
(B) 10xy
(C) 8x + y
(D) 8x - y

97. Inside a ripe bean seed, the tiny part that will grow into the roots, stem and leaves of the new plant is the:

(A) the pollen
(B) the stigma
(C) the seed coat
(D) embryo

98. The block capital letter B has how many lines of symmetry?

- **(A)** 2 lines
- **(B)** 1 vertical line
- **(C)** 0 lines
- **(D)** 1 (a horizontal line across the middle)

99. A forest gives many kinds of animals a place to live, find food and raise their young. Such a natural home is called the animal's:

- **(A)** humus
- **(B)** habitat
- **(C)** canopy
- **(D)** food chain

100. The perimeter of a rectangle with length l and breadth b can be written as the expression:

- **(A)** 4l
- **(B)** lb
- **(C)** 2l + 2b
- **(D)** l + b